

TARA CUDA USER MANUAL

MADE BY:

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Adviser:

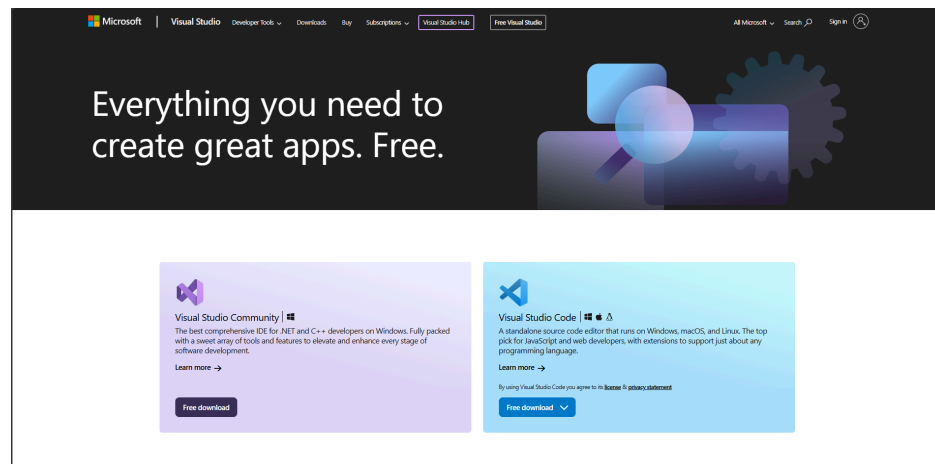
Mr. Roger Luis Uy

Specifications:

- System Units / Cloud Computing Platforms with a CUDA-capable NVIDIA Graphics Processing Unit (GPU). See the list of capable devices [here](#). The program was tested on devices with Compute Capability 7.5, 8.6, and 8.9.

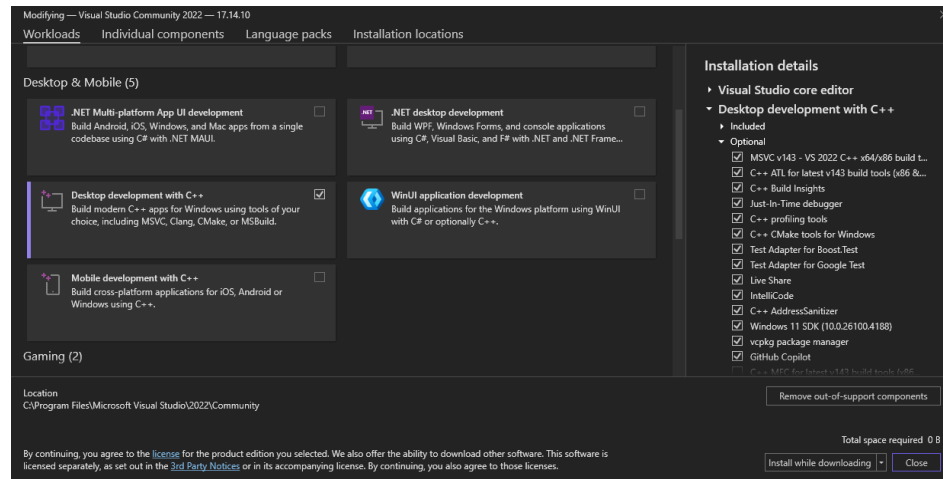
Installation:

- Visual Studio Community
 - Download Visual Studio 2022 from [Microsoft for Visual Studio Community](#).

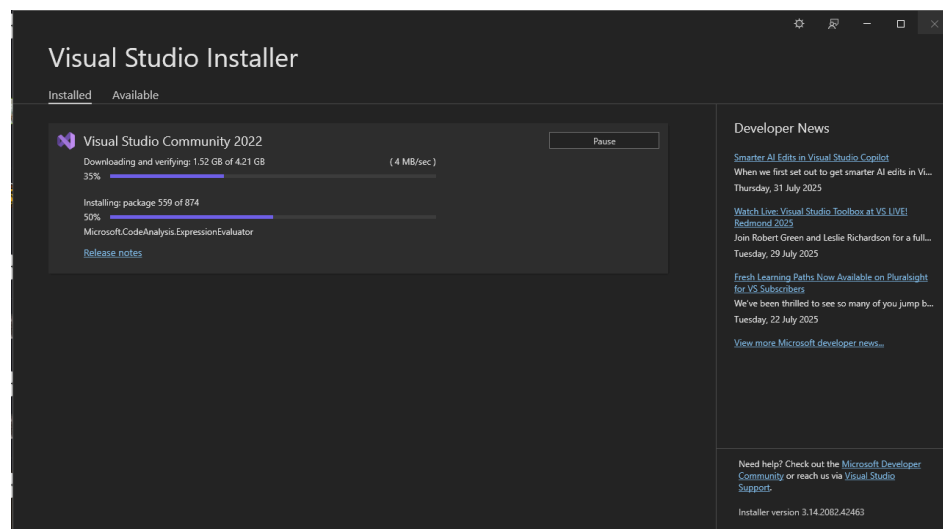


- Launch the executable file for installation and select Continue to install the Visual Studio Installer.

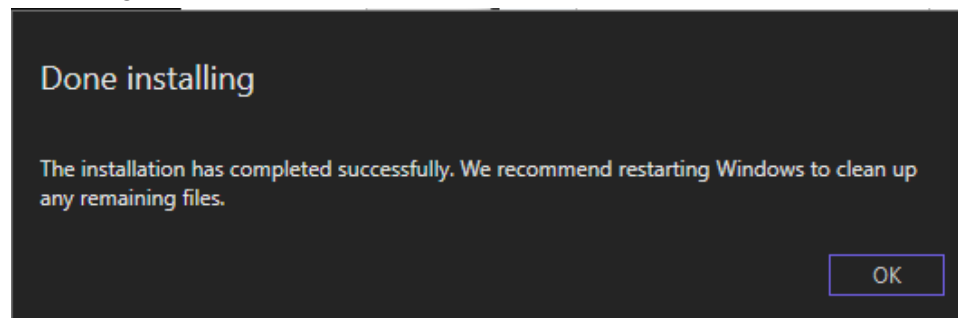
- You will be redirected to workloads, check Desktop development with C++, and press install while downloading.



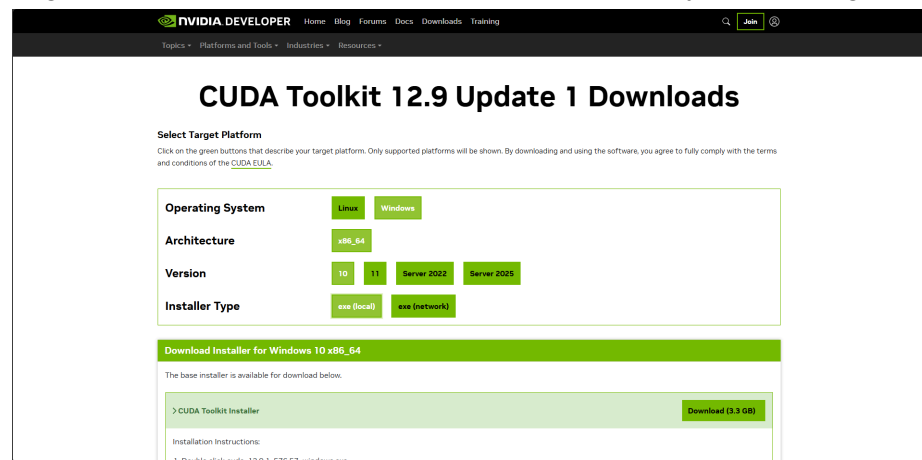
- Once accomplished, the installer will now process the installation of Visual Studio 2022.



- Wait for Visual Studio 2022 to finish installing and restart your unit to clean up remaining files.



- CUDA Toolkit
 - Download the executable file for the CUDA Toolkit from the NVIDIA Developer Page and select the options that match the platform you are using.



- Run the executable file and follow the prompts to install the package.
- Wait for the installer to finish installing the CUDA Toolkit, and now you are ready to run CUDA-specific applications.
- Program Files
 - Clone the repository from this [link](#).
 - Upon cloning the repository for TARA CUDA, select Zhang's Six Frame in CUDA.sln.

Running the Program

- Select the project and simply press the run button in Visual Studio 2022.
- Input the fasta file name for both protein and DNA. Protein first and next is DNA.

```
Input file name for protein(exclude .fasta/.fastq):  
--> testProt  
Input file name for DNA (exclude .fasta/.fastq):  
--> testDNA
```

- Choose 0 (Sequential) or 1 (CUDA) for the implementation type.

```
Choose mode [0 - Sequential] [1 - CUDA]:  
--> 1
```

- Choose 3 or 6 for the frame alignment count to be used.

```
Choose frame count [3 - 3 Frame] [6 - 6 Frame]:  
--> 6
```

- Choose between 0 or 1 to show Top 1 score or Top 5 scores

```
Show Score [0 - Top 1] [1 - Top 5]:  
--> 1
```

- After that the program will output this if you selected top 5 scores

```
24sp|A0A1P8B590|IQ019_ARATH Protein IQ-DOMAIN 19 OS=Arabidopsis thaliana OX=3702 GN=IQ019 PE=1 SV=1
Sequence: 24
DNA Input:  TGTGCCGACGTAATGTGATGATTCAAACCAGGGCGGTGTAATGGCGCTGATGATCTCAGTCAAGAGTGCTTGTCAAGCTTGGATGTTCCCGAGAGAGAGACTCAAGAATTGTTGGCTATCATAGACTTGATGTGGAATGGTTTCA
C A D V N V M I I K P G R W X W R X X S Q S R V L V S L D G S R R R E R L K N C W L S X T X C G M V S
V P T X H X X L S N Q G G G N G A D D L S Q E C L S A W M V P G E R D S R I V G Y H R L D V E W F
C R R K C D D Y Q T R A V V M A L M I S V K S A C Q L G W F P E R E T Q E L L A I I D L M W N G F
Frame Match:  M X X L S N Q G G G N G A D D S Q Q E C - - - L S A W M G S R R E R L K N C W L S X T X C G M V S
Frameshift:    * * *
Output Prot:   M Q R S S S Q L G S N T A K E S Q Q H H H H Q Y Y P W M A I K L D R - S N I S L M E S E C G S T S
Score: 67
Start to End match in Protein: 378-425
Time in ms: 1.42429

17sp|A0A1P8AUV4|MDN1_ARATH Midasin OS=Arabidopsis thaliana OX=3702 GN=MDN1 PE=1 SV=1
Sequence: 17
DNA Input:  TGTGCCGACGTAATGTGATGATTCAAACCAGGGCGGTGTAATGGCGCTGATGATCTCAGTCAAGAGTGCTTGTCAAGCTTGGATGTTCCCGAGAGAGAGACTCAAGAATTGTTGGCTATCATAGACTTGATGTGGAATGGTTTCA
C A D V N V M I I K P G R W X W R X X S Q S R V L V S L D G S R R R E R L K N C W L S X T X C G M V S
V P T X H X X L S N Q G G G N G A D D L S Q E C L S A W M V P G E R D S R I V G Y H R L D V E W F
C R R K C D D Y Q T R A V V M A L M I S V K S A C Q L G W F P E R E T Q E L L A I I D L M W N G F
Frame Match:  I I K P G R W N G A D D L S Q E V L V S - L D G S R R R D S R N C W Y I D L M W N
Frameshift:    * * *
Output Prot:   L L D P D L M D G A S D M R I E V L V S A L D A E Q L R D - R S S W Y V - V W W N
Score: 65
Start to End match in Protein: 3970-4008
Time in ms: 13.2897

23sp|A0A1P8B554|THLP1_ARATH Thaumatin-like protein 1 OS=Arabidopsis thaliana OX=3702 GN=THLP1 PE=2 SV=1
Sequence: 23
DNA Input:  TGTGCCGACGTAATGTGATGATTCAAACCAGGGCGGTGTAATGGCGCTGATGATCTCAGTCAAGAGTGCTTGTCAAGCTTGGATGTTCCCGAGAGAGAGACTCAAGAATTGTTGGCTATCATAGACTTGATGTGGAATGGTTTCA
C A D V N V M I I K P G R W X W R X X S Q S R V L V S L D G S R R R E R L K N C W L S X T X C G M V S
V P T X H X X L S N Q G G G N G A D D L S Q E C L S A W M V P G E R D S R I V G Y H R L D V E W F
C R R K C D D Y Q T R A V V M A L M I S V K S A C Q L G W F P E R E T Q E L L A I I D L M W N G F
Frame Match:  R K C D D Y Q T R A V G N G A D D S V K S A C Q L D G S P E E R L K N C W L S X T X C - - G M V S
Frameshift:    * * *
Output Prot:   Q K C P T - E L R - F G S G S - - A C K S A C E A F G S P E Y C C S G A V A S P T E C K P S M Y S
Score: 63
Start to End match in Protein: 177-221
Time in ms: 0.968096

46sp|A05VXB|DOG1_ARATH Protein DELAY OF GERMINATION 1 OS=Arabidopsis thaliana OX=3702 GN=DOG1 PE=1 SV=1
Sequence: 46
DNA Input:  TGTGCCGACGTAATGTGATGATTCAAACCAGGGCGGTGTAATGGCGCTGATGATCTCAGTCAAGAGTGCTTGTCAAGCTTGGATGTTCCCGAGAGAGAGACTCAAGAATTGTTGGCTATCATAGACTTGATGTGGAATGGTTTCA
C A D V N V M I I K P G R W X W R X X S Q S R V L V S L D G S R R E R L K N C W L S X T X C G M V S
V P T X H X X L S N Q G G G N G A D D L S Q E C L S A W M V P G E R D S R I V G Y H R L D V E W F
C R R K C D D Y Q T R A V V M A L M I S V K S A C Q L G W F P E R E T Q E L L A I I D L M W N G F
Frame Match:  R N V D D Y Q N Q G G G N G A D - - D L S Q E C L S
Frameshift:    * * *
Output Prot:   R N I D G Y E S S G G G G G A S L S D L S A E Q L A
Score: 61
Start to End match in Protein: 122-147
Time in ms: 1.0385

45sp|A0WF59|CACLC_ARATH Anoctamin-like protein At1g73020 OS=Arabidopsis thaliana OX=3702 GN=At1g73020 PE=1 SV=1
Sequence: 45
DNA Input:  TGTGCCGACGTAATGTGATGATTCAAACCAGGGCGGTGTAATGGCGCTGATGATCTCAGTCAAGAGTGCTTGTCAAGCTTGGATGTTCCCGAGAGAGAGACTCAAGAATTGTTGGCTATCATAGACTTGATGTGGAATGGTTTCA
C A D V N V M I I K P G R W X W R X X S Q S R V L V S L D G S R R E R L K N C W L S X T X C G M V S
V P T X H X X L S N Q G G G N G A D D L S Q E C L S A W M V P G E R D S R I V G Y H R L D V E W F
C R R K C D D Y Q T R A V V M A L M I S V K S A C Q L G W F P E R E T Q E L L A I I D L M W N G F
Frame Match:  R R N V M I I K P G R W X H A L M I S V K S A C Q - L G - - W - - - - F P R E R L K N V G Y H R L - - - D V - - - E W F
Frameshift:    * * *
Output Prot:   R K N A A L L - - A R W Q I N C L V G P S Q G Y R F L G N E W S S L P F P K E L I K N L G N E R A K E K E A Y Q R Y E W F
Score: 60
Start to End match in Protein: 253-311
Time in ms: 1.91533
```

Understanding the Output

```
24sp|A0A1P8B590|IQ019_ARATH Protein IQ-DOMAIN 19 OS=Arabidopsis thaliana OX=3702 GN=IQD19 PE=1 SV=1
Sequence: 24
DNA Input:  TGTGCCGACGTAATGTGATGATTATCAAAACAGGGCGGTGTAATGGCGCTGATCTCAGTCAAGAGTGCTTGTCAAGCTTGGATGGTCCCGGAGAGAGAGACTCAAGAATTGTTGGCTATCATAGACTTGATGTGGAATGGTTTCA
            C A D V N V M I I K P G R W X W R X X S Q S R V L V S L D G S R R E R L K N C W L S X T X C G M V S
            V P T X M X X L S N Q G G G N G A D D L S Q E C L S A W M V P G E R D S R I V G Y H R L D V E W F
            C R R K C D D Y Q T R A V V M A L M I S V K S A C Q L G W F P E R E T Q E L L A I I D L M W N G F
Frame Match: M X X L S N Q G G G N G A D D S Q Q E C - - - L S A W M G S R R E R L K N C W L S X T X C G M V S
Frameshift:  * * *
Output Prot:  M Q R S S S Q L G S N T A K E S Q Q H H H H Q Y Y P W M A I K L D R - S N I S L M E S E C G S T S
Score: 67
Start to End match in Protein: 378-425
Time in ms: 1.42429
```

- The first line shows the protein name, and the second line shows the protein number of the fasta file.
- The fourth line shows the DNA input/query, and the fifth to seventh lines are the frames it can create using the DNA.

```
DNA Input:  TGTGCCGACGTAATGTGATGATTATCAAAACAGGGCGGTGTAATGGCGCTGATCTCAGTCAAGAGTGCTTGTCAAGCTTGGATGGTCCCGGAGAGAGAGACTCAAGAATTGTTGGCTATCATAGACTTGATGTGGAATGGTTTCA
            C A D V N V M I I K P G R W X W R X X S Q S R V L V S L D G S R R E R L K N C W L S X T X C G M V S
            V P T X M X X L S N Q G G G N G A D D L S Q E C L S A W M V P G E R D S R I V G Y H R L D V E W F
            C R R K C D D Y Q T R A V V M A L M I S V K S A C Q L G W F P E R E T Q E L L A I I D L M W N G F
```

- Ninth to eleventh lines show the Frame match where the protein matches from the frames created from the DNA input, Frameshift shows (*) where a frameshift occurs, and Output Prot shows the actual protein that matched to the frame shift alignment.

```
Frame Match: M X X L S N Q G G G N G A D D S Q Q E C - - - L S A W M G S R R E R L K N C W L S X T X C G M V S
Frameshift:  * * *
Output Prot:  M Q R S S S Q L G S N T A K E S Q Q H H H H Q Y Y P W M A I K L D R - S N I S L M E S E C G S T S
```

- The Thirteenth and Fourteenth lines show the score from the DNA-Protein Alignment and the Protein indexes where it matched from the DNA.

```
Score: 67
Start to End match in Protein: 378-425
```

- The final line shows the amount of time taken by the program to perform the matrix scoring for the given alignment.

```
Time in ms: 1.42429
```